Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/17/2005
PI Name:	Baker, John Ph.D.		
Project Title:	Magnetically-Assisted Combustion Experiment (MACE)		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	35487-0276	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	05/07/2002	**End Date:**	05/06/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	2
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Saito, Kozo (University of Kentucky)		
Grant/Contract No.:	NCC3-964		
Performance Goal No.:			
Performance Goal Text:			
	This investigation examines the interaction between magnetic fields and combustion processes. Using both experimental and mathematical models, the impact of magnetic fields on laminar diffusion flames will be quantified. Initially, terrestrial based experiments will be conducted to gain insight into the magnetocombustion interaction. This data will also be used to gauge the effectiveness of mathematical magnetocombustion models. The mathematical models will then be used to guide the development of a series of reduced gravity experiments. Unfortunately, due to the fact that both magnetic and gravitational forces are body forces, it is impossible to isolate magnetic field effects in a terrestrial environment. The experimental component of the study will examine the impact of non-uniform magnetic fields on laminar diffusion flames in a reduced gravity environment where the impact of gravitational acceleration is greatly reduced. The goal of the study is to explore the interaction between magnetic fields and combustion processes. The three principal objectives of the research are:		

Task Description:	1. To experimentally observe, in an environment where buoyant forces are negligible, the effect of non-uniform magnetic fields on diffusion flame characteristics and on the associated temperature fields; 2. To better understand the behavior of laminar diffusion flames in non-uniform magnetic fields and the impact that paramagnetism and diamagnetism have with regard to transport phenomena in the vicinity of such flames; and 3. To formulate mathematical models that can accurately predict experimentally observed magnetocombustion behavior. Magnetic fields affect diffusion flame behavior as a result of the paramagnetic and diamagnetic characteristics of the constituent gases. By developing a fundamental understanding of this interaction, it will be possible to intelligently control the transport characteristics in the vicinity of the flame. This study seeks to quantify magnetocombustion interactions by examining such behavior in an environment where magnetic fields effects can be isolated from buoyancy effects.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By examining the impact of magnetic fields on laminar diffusion flames, a better fundamental understanding of how magnetic fields can be used to intelligently control the behavior of flame is being developed. This understanding could potentially be used to develop technology which can improve combustion efficiency and reduce harmful emissions from combustion related processes.
Task Progress:	The past year has primarily focused on the design and fabrication of an experimental test cell for the MACE project. A first draft of the TEDP was received by NASA and significant revisions to document and to the design of the test cell were required. This has resulted in the inclusion of a much expanded structural analysis in the TEDP. While the initial design of the experimental frame met or exceeded NASA requirements, complete documentation of the calculations used to show that these requirements were met have now been included in the TEDP. As per other comments made regarding the first version of the TEDP, the overall design and arrangement of the experimental components have been modified. This redesign and refabrication took considerable time as it required a dismantling of the original test cell which had flown previously during two NASA Reduced Gravity Student Flight Opportunities Program campaigns. In parallel to the experimental effort, the effort related to the theoretical modeling of magnetocombustion behavior continued. A complete investigation as to the impact of non-uniform magnetic fields on laminar diffusion flames was conducted using the magnetocombustion flame height model previously developed. In addition, the flame height model is being modified to predict flame shape. Data from the theoretical model have not only yielded insight into the fundamental nature of the interaction between magnetic fields and diffusion flames but have also been useful in the redesign of the experimental test cell. The second draft of the TEDP was recently submitted to NASA and comments/suggestions regarding the second TEDP have been received from NASA.
Bibliography Type:	Description: (Last Updated: 01/17/2005)
Abstracts for Journals and Proceedings	Raju, S.S.; Baker, J.; and Calvert, M.E "Theoretical Model of Laminar Diffusion Flame Height in Non-Uniform Magnetic Field" Southeastern Conference on Theoretical and Applied Mechanics Developments in Theoretical and Applied Mechanics, Mahfuz, M. and Hosur, M.V., ed, Vol. XXII, pp. 519-528. , Aug-2004
Articles in Peer-reviewed Journals	Baker, J.; Srireddy, M.; Varagani, R. "Buoyancy-Controlled Laminar Diffusion Slot Flame Heights: A Comparison of Theoretical Predictions and Microgravity Results" Microgravity Science and Technology, Vol. XIV, No. 4, pp.27-35 , Dec-2003
Articles in Peer-reviewed Journals	Baker, J.; Calvert, M.E. "A Study of the Characteristics of Slotted Laminar Jet Diffusion Flames in the Presence of Non-Uniform Magnetic Fields" Combustion and Flame, Vol. 133, No.3, pp.345-357 , Dec-2003